

BrainChart allows users to compute (per)centile scores using an out-of-sample maximum likelihood estimation against the bootstrapped model parameters obtained from a large, aggregated neuroimaging dataset (see <https://www.nature.com/articles/s41586-022-04554-y>). No uploaded data is stored in the app, though if you want your dataset to be included in the next iteration of the normative models (and to become a member of the BrainChart working group and consortium) please contact us (BrainCharts@gmail.com). To use the app, please **download the included template** and make sure your data is formatted in the same way, adhering to the following guidance for each column:

- **Participant:** a unique participant identifier as that used in your study
- **Age:** the participants age in years
- **age_days:** the participants age in post-conception days, which can be roughly estimated as: $(\text{years} * 365.245) + 280$
- **sex:** a binary factor identifying “Male” or “Female”
- **study:** the name of your study. Even if working with a dataset included in our models, avoid using the same study name/acronym (see app or paper) when using this out-of-sample tool.
- **fs_version:** the FreeSurfer version used to obtain gross volumetric measures, the currently included options are listed in the template, if your version is not included please code this field as NA and the parametrisation will default to the average across included version. This will not affect the estimate as long as your data was uniformly processed, since this offset would then be perfectly orthogonal to a study specific effect.
- **country:** optional country argument, leave as NA if not known
- **run:** run refers to a scenario where a participant will have multiple measures within the same session. For example multiple MPRAGE scans within the same scanning session. Please encode as numeric (e.g. 1, 2, ... n)
- **session:** refers to possibly multiple scanning sessions at different time-points. Please encode as numeric (e.g. 1, 2, ... n)
- **dx:** refers to diagnosis, no specific classification system is used for clinical cohorts, but the label “CN” is used for controls. The estimator works most accurately on sample that contain 100 or more CN participants
- **GMV:** refers to cortical grey matter volume and is analogous with freesurfers “Total cortical grey matter volume” label in the aseg.stats file
- **WMV:** refers to cerebral white matter volume and is analogous with freesurfers “Total cerebral white matter volume” label in the aseg.stats file
- **sGMV:** refers to subcortical grey matter volume and is analogous with freesurfers “Subcortical grey matter volume” label in the aseg.stats file
- **Ventricles:** refers to ventricular volume and is analogous with freesurfers “Volume of ventricles and choroid plexus” label in the aseg.stats file

The user can leave any additional fields relevant for their own study. Parameter estimates and centile scores will be appended to the original upload. **For clarity, centile scores are only computed for the selected brain feature. Therefore, repeat for each requested feature.**